

5 Trigonometry	5.1	Understand and use the definitions of sine, cosine and tangent for all arguments; the sine and cosine rules; the area of a triangle in the form $\frac{1}{2}ab \sin C$ Work with radian measure, including use for arc length and area of sector.	Use of x and y coordinates of points on the unit circle to give cosine and sine respectively, including the ambiguous case of the sine rule. Use of the formulae $s = r\theta$ and $A = \frac{1}{2}r^2\theta$ for arc lengths and areas of sectors of a circle.
	5.2	Understand and use the standard small angle approximations of sine, cosine and tangent $\sin \theta \approx \theta$, $\cos \theta \approx 1 - \frac{\theta^2}{2}$, $\tan \theta \approx \theta$ Where θ is in radians.	Students should be able to approximate, e.g. $\frac{\cos 3x - 1}{x \sin 4x}$ when x is small, to $-\frac{9}{8}$
	5.3	Understand and use the sine, cosine and tangent functions; their graphs, symmetries and periodicity. Know and use exact values of sin and cos for $0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}, \pi$ and multiples thereof, and exact values of tan for $0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \pi$ and multiples thereof.	Knowledge of graphs of curves with equations such as $y = \sin x$, $y = \cos(x + 30^\circ)$, $y = \tan 2x$ is expected.
	5.4	Understand and use the definitions of secant, cosecant and cotangent and of arcsin, arccos and arctan; their relationships to sine, cosine and tangent; understanding of their graphs; their ranges and domains.	Angles measured in both degrees and radians.

Topics	What students need to learn:		
	Content		Guidance
5 Trigonometry <i>continued</i>	5.5	Understand and use $\tan \theta = \frac{\sin \theta}{\cos \theta}$ Understand and use $\sin^2 \theta + \cos^2 \theta = 1$ $\sec^2 \theta = 1 + \tan^2 \theta$ and $\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta$	These identities may be used to solve trigonometric equations and angles may be in degrees or radians. They may also be used to prove further identities.
	5.6	Understand and use double angle formulae; use of formulae for $\sin (A \pm B)$, $\cos (A \pm B)$, and $\tan (A \pm B)$, understand geometrical proofs of these formulae. Understand and use expressions for $a \cos \theta + b \sin \theta$ in the equivalent forms of $r \cos (\theta \pm \alpha)$ or $r \sin (\theta \pm \alpha)$	To include application to half angles. Knowledge of the $\tan(\frac{1}{2}\theta)$ formulae will <i>not</i> be required. Students should be able to solve equations such as $a \cos \theta + b \sin \theta = c$ in a given interval.
	5.7	Solve simple trigonometric equations in a given interval, including quadratic equations in sin, cos and tan and equations involving multiples of the unknown angle.	Students should be able to solve equations such as $\sin (x + 70^\circ) = 0.5$ for $0 < x < 360^\circ$, $3 + 5 \cos 2x = 1$ for $-180^\circ < x < 180^\circ$ $6 \cos^2 x + \sin x - 5 = 0$, $0 \leq x < 360^\circ$ These may be in degrees or radians and this will be specified in the question.
	5.8	Construct proofs involving trigonometric functions and identities.	Students need to prove identities such as $\cos x \cos 2x + \sin x \sin 2x \equiv \cos x$.
	5.9	Use trigonometric functions to solve problems in context, including problems involving vectors, kinematics and forces.	Problems could involve (for example) wave motion, the height of a point on a vertical circular wheel, or the hours of sunlight throughout the year. Angles may be measured in degrees or in radians.